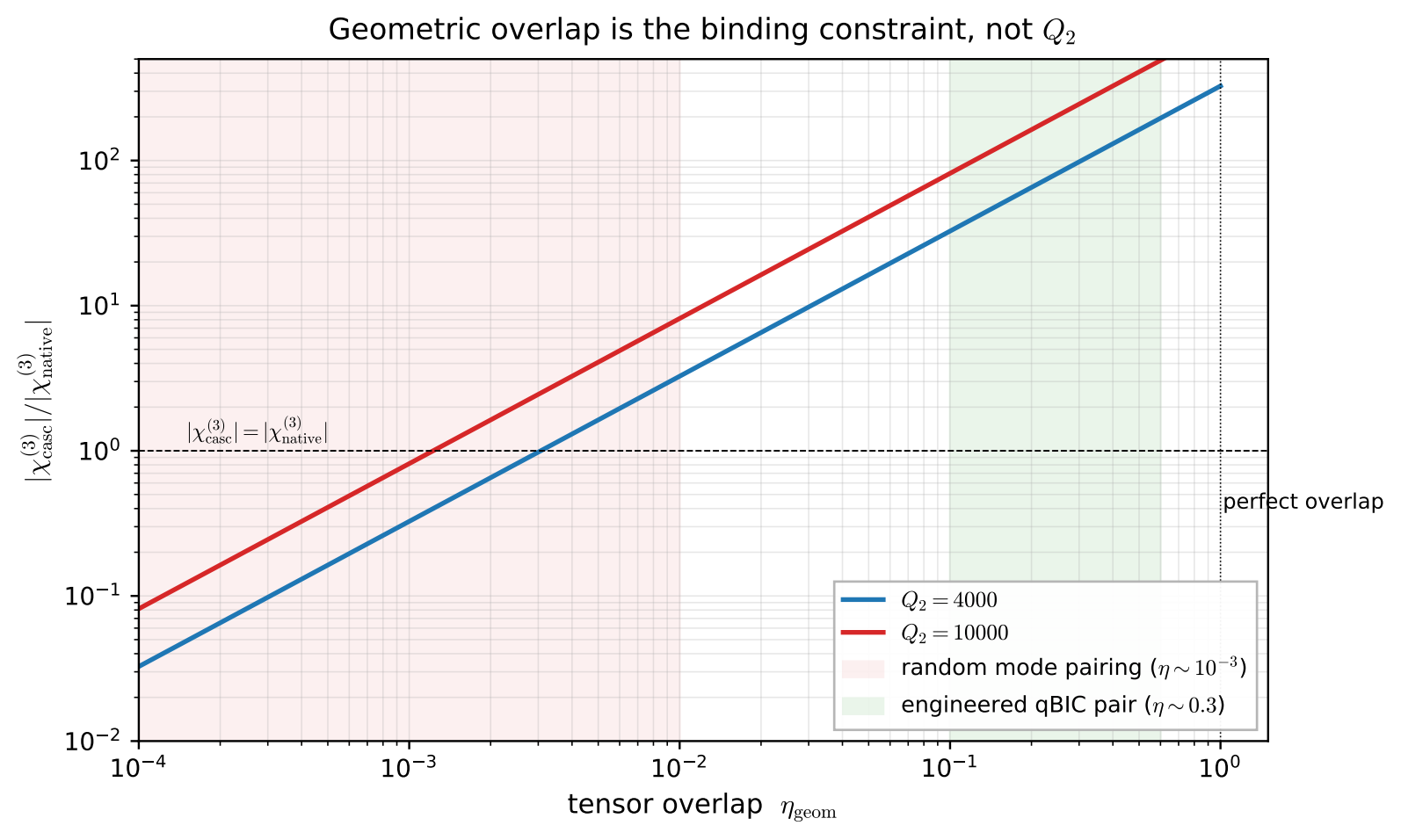
Geometric overlap is the binding constraint, not Q_2
$|\chi^{(3)}_{\mathrm{casc}}| / |\chi^{(3)}_{\mathrm{native}}|$
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
$|\chi^{(3)}_{\mathrm{casc}}| = |\chi^{(3)}_{\mathrm{native}}|$
perfect overlap
$Q_2 = 4000$
$Q_2 = 10000$
random mode pairing ($\eta \sim 10^{-3}$)
engineered qBIC pair ($\eta \sim 0.3$)
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
tensor overlap η_{geom}